7 Fig. 7.1 shows two charged metal plates. X marks the position of the centre of the space between the plates.

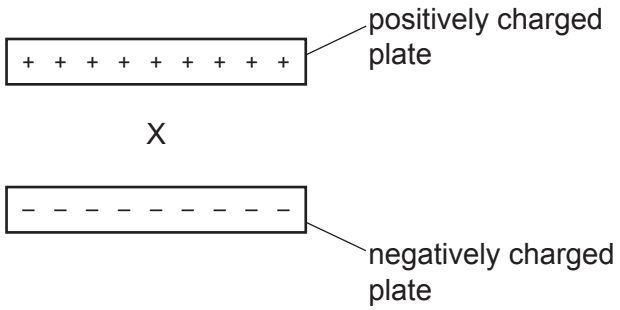


Fig. 7.1

(a) (i) On Fig. 7.1, draw at least **four** field lines to show the pattern and the direction of the electric field between the two charged plates. [2]

(ii) Describe the effect on a negatively charged particle placed at X.

.....

..... [1]

(b) During a thunderstorm, an electric field is set up between a cloud and the ground. Charges on the cloud and on the ground are shown in Fig. 7.2.

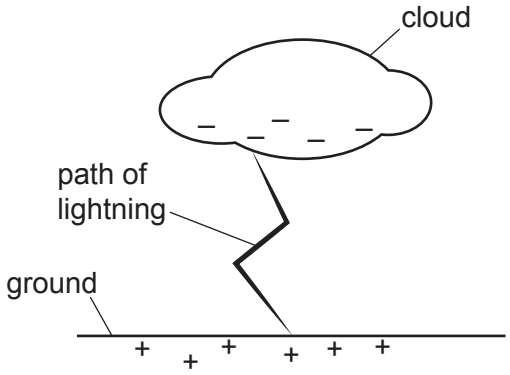


Fig. 7.2

The lightning shown in Fig. 7.2 discharges a current of 28 000 A for 0.0012 s.

(i) Calculate the charge that flows from the cloud to the ground.

charge = ... [2]

(ii) The lightning transfers 1.2×10^8 J of energy.

Calculate the potential difference between the base of the cloud and the ground.

potential difference = ... [2]

[Total: 7]